

■ Python Scripting

Phase 1 — Complete Learning Path

What to Learn · Where to Learn · What to Practice

Duration: 4 – 6 Weeks | Goal: Freelance-Ready Foundations

■ PHASE OVERVIEW

Phase 1 builds the mental model and muscle memory you need before writing any automation scripts. Every topic here maps directly to a real freelancing task you will do on Fiverr. Skipping this phase means your scripts will be brittle and hard to sell. Master it and everything after becomes 3× easier.

End goal: After 6 weeks you can read any beginner Python script, understand what it does, and modify it for a client.

■ Duration	4 – 6 Weeks
■ Topics	8 Core Areas
■ Projects	6 Mini Projects
■ Resources	5 Free Platforms

■ Topic 01 — Setup & Environment

Day 1–2

■ WHY THIS MATTERS

You can't code without a working environment. Getting this right once saves hours of frustration.

■ WHAT TO LEARN

- Install Python 3.11+ from python.org
- Install VS Code + Python extension (your main editor)
- Understand how to run a .py file from terminal
- Learn what pip is and how to install packages
- Create your first 'hello.py' and run it

■ WHAT TO PRACTICE

- ✓ Install Python and verify with: `python --version`
- ✓ Write and run a file that prints your name and today's date
- ✓ Install the 'requests' library with pip and verify

■ RESOURCES

python.org/downloads

Official installer

code.visualstudio.com

VS Code editor

YouTube: 'Python installation 2024'

Visual guide

■ *On Windows, tick 'Add Python to PATH' during installation — skip this and nothing works.*

■ Topic 02 — Variables & Data Types

Day 3–5

■ WHY THIS MATTERS

Variables are containers. Data types tell Python what kind of data you are storing. Every single script uses these.

■ WHAT TO LEARN

- String → text in quotes: `name = 'Umar'`
- Integer → whole numbers: `age = 22`
- Float → decimals: `price = 4.99`
- Boolean → True / False: `is_active = True`
- Type conversion: `int('5')`, `str(100)`, `float('3.14')`
- f-strings for formatting: `f'Hello {name}'`

■ WHAT TO PRACTICE

- ✓ Create variables for: your name, age, city, hourly rate
- ✓ Print a sentence using all four using an f-string
- ✓ Convert a string '42' to int and multiply it by 3
- ✓ Mini project: Build a 'profile card' that prints formatted user info

■ RESOURCES

[w3schools.com/python/python_variables.asp](https://www.w3schools.com/python/python_variables.asp)

Interactive lessons

replit.com

Code in browser, no install needed

■ *f-strings (`f'Hello {name}'`) are your best friend. Use them everywhere — they are the modern Python way.*

■ Topic 03 — Conditionals & Loops

Day 6–9

■ WHY THIS MATTERS

Conditionals make decisions. Loops repeat actions. Together they are the engine of every automation script you will write.

■ WHAT TO LEARN

- if / elif / else statements
- Comparison operators: ==, !=, >, <, >=, <=
- Logical operators: and, or, not
- for loop — iterate over a range or list
- while loop — repeat until a condition is false
- break and continue — control loop flow
- range(start, stop, step)

■ WHAT TO PRACTICE

- ✓ Write a grade checker: if score > 90 → 'A', etc.
- ✓ Print all even numbers from 1 to 100 using a loop
- ✓ Write a loop that counts down from 10 to 1
- ✓ Mini project: FizzBuzz (classic coding interview problem)
- ✓ Mini project: Simple number guessing game with while loop

■ RESOURCES

cs50.harvard.edu/python

Free Harvard Python course — best for logic

w3schools.com/python/python_conditions.asp

Quick reference

■ *FizzBuzz is famous because it tests if you truly understand conditionals + loops. If you can build it, you are solid.*

■ Topic 04 — Lists, Tuples & Dictionaries

Day 10–13

■ WHY THIS MATTERS

Real-world data comes in collections — product lists, user records, API responses. These structures hold that data.

■ WHAT TO LEARN

- List: ordered, changeable → `fruits = ['apple', 'mango']`
- Tuple: ordered, unchangeable → `coords = (24.8, 67.0)`
- Dictionary: key-value pairs → `user = {'name': 'Umar', 'age': 22}`
- List methods: `.append()`, `.remove()`, `.sort()`, `.pop()`
- Dict methods: `.get()`, `.keys()`, `.values()`, `.items()`
- Looping over lists and dictionaries
- List comprehensions: `[x*2 for x in numbers]`

■ WHAT TO PRACTICE

- ✓ Create a list of 5 tasks, print them numbered (1. Task, 2. Task...)
- ✓ Create a dictionary for a Fiverr gig (title, price, delivery_days)
- ✓ Filter a list of numbers — keep only those above 50
- ✓ Mini project: Simple contact book using a dictionary

■ RESOURCES

automatetheboringstuff.com — Chapter 4 & 5

Best practical coverage

pythontutor.com

Visualise what happens in memory — mind-blowing

■ *Dictionaries are the closest thing to real API data (JSON). Get comfortable with them — you will use them constantly.*

■ WHY THIS MATTERS

Functions let you write code once and reuse it. Every professional script is a collection of well-named functions.

■ WHAT TO LEARN

- `def` keyword — defining a function
- Parameters and arguments
- `return` statement — getting values back
- Default parameter values
- Keyword arguments
- `*args` and `**kwargs` (basic understanding)
- Scope: local vs global variables
- Docstrings — documenting your function

■ WHAT TO PRACTICE

- ✓ Write a function that calculates Fiverr earnings after 20% fee
- ✓ Write a function that greets a user by name (with default 'friend')
- ✓ Refactor your earlier mini projects to use functions
- ✓ Mini project: BMI calculator with a `calculate_bmi(weight, height)` function

■ RESOURCES

[cs50.harvard.edu/python — Week 2](https://cs50.harvard.edu/python/Week2/)

Functions explained brilliantly

realpython.com/defining-your-own-python-function

Deep article

■ *Rule of thumb: if you copy-paste code twice, it should be a function. This habit separates beginners from developers.*

■ Topic 06 — File Handling

Day 18–21

■ WHY THIS MATTERS

Clients pay for automation that reads and writes files — renaming folders, extracting data from CSVs, generating reports. This is Fiverr gold.

■ WHAT TO LEARN

- `open()` — the built-in file function
- Modes: 'r' read, 'w' write, 'a' append, 'rb' binary
- `with` statement — proper file handling
- Reading line by line vs all at once
- Writing and appending to files
- Working with CSV files using the `csv` module
- `os` module basics: listing files, creating directories

■ WHAT TO PRACTICE

- ✓ Write a script that reads a `.txt` file and counts the words
- ✓ Write a script that creates a daily log file with timestamp
- ✓ Read a CSV of names, print each one with a greeting
- ✓ Mini project: Rename all files in a folder by adding a prefix

■ RESOURCES

automatetheboringstuff.com — Chapter 8 & 9

The best coverage of this topic

docs.python.org/3/tutorial/inputoutput.html

Official docs

■ *Always use the 'with open()' pattern. It automatically closes the file even if your code crashes.*

■ Topic 07 — Error Handling

Day 22–24

■ WHY THIS MATTERS

Beginner scripts crash. Professional scripts handle errors gracefully. Clients notice the difference immediately.

■ WHAT TO LEARN

- try / except block
- Common exceptions: FileNotFoundError, ValueError, KeyError, IndexError
- except Exception as e — catching the error message
- finally block — runs no matter what
- raise — deliberately raising an error
- Custom error messages for the user

■ WHAT TO PRACTICE

- ✓ Wrap a file-open script in try/except — handle missing file
- ✓ Wrap int(input()) in try/except to handle non-number input
- ✓ Add error handling to your CSV reader from Topic 6
- ✓ Mini project: A safe calculator that never crashes

■ RESOURCES

realpython.com/python-exceptions

Best explanation online

w3schools.com/python/python_try_except.asp

Quick reference

■ *On Fiverr, clients will try your script with wrong inputs. Scripts that crash = bad reviews. Error handling = 5-star reviews.*

■ Topic 08 — Modules & Libraries

Day 25–28

■ WHY THIS MATTERS

Python's superpower is its library ecosystem. You never build from scratch — you import and extend.

■ WHAT TO LEARN

- import statement and from...import syntax
- Standard library: os, sys, datetime, math, random, json
- Third-party libraries: installing with pip
- datetime — working with dates and times
- json — parsing and creating JSON data
- random — generating random data (useful for testing)
- Creating your own module (your own .py file imported elsewhere)

■ WHAT TO PRACTICE

- ✓ Print today's date formatted as: 'Monday, 26 June 2026'
- ✓ Write a script that generates 5 random passwords
- ✓ Parse a JSON string and extract specific values
- ✓ Mini project: A mini CLI tool that uses 3+ modules together

■ RESOURCES

docs.python.org/3/library

Official standard library reference

pypi.org

Browse all third-party packages

automatetheboringstuff.com — Chapter 17

Datetime coverage

■ *JSON is the language of the internet — APIs return JSON, configs are JSON. Mastering json module = half of API scripting done.*

■ 6-Week Daily Schedule

Follow this schedule consistently — 1.5 to 2 hours per day is enough. Weekends are for building mini projects and reviewing the week.

Week 1	Focus
Mon–Tue	Setup environment, install Python & VS Code, run first file
Wed–Thu	Variables, data types, f-strings — build profile card
Fri	Review + push code to GitHub
Sat–Sun	Mini project: Personal info printer with all data types

Week 2	Focus
Mon–Tue	Conditionals: if/elif/else, comparison & logical operators
Wed–Thu	Loops: for, while, break, continue, range()
Fri	Review + FizzBuzz challenge
Sat–Sun	Mini project: Number guessing game

Week 3	Focus
Mon–Tue	Lists — create, loop, modify, comprehensions
Wed–Thu	Dictionaries — key-value, looping, nested dicts
Fri	Tuples + practice mixing all three
Sat–Sun	Mini project: Contact book with add/search/delete

Week 4	Focus
Mon–Tue	Functions — define, parameters, return, scope
Wed–Thu	Default args, *args, **kwargs, docstrings
Fri	Refactor all previous mini projects using functions
Sat–Sun	Mini project: BMI calculator + Fiverr fee calculator

Week 5	Focus
Mon–Tue	File handling — open, read, write, append
Wed–Thu	CSV files + os module basics
Fri	Error handling — try/except/finally

Sat–Sun	Mini project: File renamer + word count tool
Week 6	Focus
Mon–Tue	Modules — datetime, json, random, math
Wed–Thu	Build a mini CLI tool combining everything
Fri	Polish all 6 projects, write READMEs
Sat–Sun	Push everything to GitHub — your Phase 1 portfolio is done!

■ Master Resource List

automatetheboringstuff.com

FREE BOOK

The single best free resource for learning Python through real automation tasks. Every chapter teaches a skill you can sell on Fiverr.

cs50.harvard.edu/python

FREE COURSE

Harvard's CS50P course — free, world-class, beginner-friendly. Covers all of Phase 1 in video format with graded problem sets.

replit.com

ONLINE IDE

Code Python in your browser — no installation needed. Great for quick practice when you are on mobile or a slow machine.

pythontutor.com

VISUALISER

Paste your code and watch it execute step by step. Essential for understanding loops, functions and scope visually.

w3schools.com/python

REFERENCE

Quick syntax reference for any topic. Not for deep learning, but perfect when you forget how something works mid-project.

■ GitHub Portfolio — Phase 1 Structure

Every mini project you build goes on GitHub. Structure your repo like this so clients and employers can navigate it easily:

```
04_functions/
```

■ Every README.md should include:

- What the script does (one sentence)
- How to run it: `python filename.py`
- What input it expects
- Example output (paste a screenshot or text output)

■ *Tip: Commit after every session with a clear message like 'Add fizzbuzz solution with comments'. This shows consistent progress to any client who checks your profile.*

python.

You are building a product — your skill set — that you will sell on your own. This is a significant investment. Complete Phase 1 and you will have 6 real projects and the foundations to automate anything.